IdiomX: A Bilingual Benchmark for Idiom Understanding, Generation, and Cross-Lingual Retrieval

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Abstract

Idiomatic expressions remain a major challenge in natural language processing due to their non-compositional meanings, strong dependence on context, and difficulty in cross-lingual alignment. Existing idiom datasets are often limited in scale, contextual diversity, or multilingual coverage, which restricts their usefulness for modern language models.

In this paper, we introduce IdiomX, a large-scale bilingual (English–Arabic) benchmark for idiom understanding and cross-lingual semantic retrieval. IdiomX is constructed through a reproducible multi-stage pipeline that combines lexical resource extraction, controlled large language model (LLM) enrichment, and structured validation to ensure high-quality and semantically consistent data.

The dataset contains 174,956 contextualized examples covering 12,823 English idioms, with aligned English–Arabic meanings, idiomatic and literal usage labels, and rich linguistic metadata. Additional quality controls include deduplication, semantic validation, and leakage-safe splitting to ensure reliable evaluation.

We evaluate IdiomX across three benchmark tasks: idiom detection, context-to-idiom prediction, and Arabic-context-to-English idiom retrieval. For idiom detection, transformer-based models significantly outperform lexical baselines, with RoBERTa achieving 93.06% accuracy compared to 87.32% for TF-IDF + Logistic Regression. For context-to-idiom prediction, a hybrid retrieval approach reaches 0.867 Top-1 accuracy and 0.894 MRR. In the cross-lingual setting, fine-tuned multilingual embedding models achieve 64.86% Top-1 and 81.46% Top-5 accuracy.

These results demonstrate that contextual modeling is essential for understanding figurative language and that IdiomX provides a strong, reusable benchmark for studying idiom detection, semantic retrieval, and cross-lingual language understanding.

The proposed pipeline is modular and can be extended to additional languages, enabling scalable multilingual idiom benchmarking.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. Introduction

Idiomatic expressions are a well-known challenge for natural language understanding because their meanings are often non-compositional: they cannot be inferred directly from the meanings of their individual words. Expressions such as *kick the bucket* or *spill the beans* require contextual and cultural interpretation rather than literal reading. As a result, idioms remain difficult for computational models in tasks such as semantic interpretation, machine translation, information retrieval, and contextual reasoning.

Despite the significant progress achieved by large language models and contextual embeddings, idiomatic language continues to be a frequent source of error. While modern models are capable of capturing general linguistic patterns, they often struggle to distinguish between literal and figurative usage when context is subtle or ambiguous. One key limitation is the availability of suitable datasets. Existing idiom resources are often relatively small, focused on a single task, or primarily monolingual. Datasets such as VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4] have made important contributions, but they do not fully address the need for large-scale, context-rich, and bilingual benchmarks that support both monolingual and cross-lingual evaluation.

To address this gap, we introduce IdiomX, a large-scale bilingual dataset for idiom understanding and cross-lingual semantic retrieval. IdiomX contains 174,956 contextualized examples covering 12,823 English idioms, with aligned English–Arabic meanings, idiomatic and literal usage labels, and additional linguistic metadata. Each idiom is represented across multiple contexts, improving both diversity and evaluation realism.

A key strength of IdiomX is its structured and reproducible construction pipeline. The dataset combines curated idiom resources with controlled large language model (LLM) enrichment and validation. This approach allows us to significantly expand coverage while maintaining semantic consistency and data quality. The pipeline is modular and can be extended to additional languages with minimal changes.

IdiomX supports three benchmark tasks: idiom detection, context-to-idiom retrieval, and Arabic-to-English idiom retrieval. These tasks provide a unified framework for evaluating figurative language understanding across both monolingual and multilingual settings. Experimental results show that contextual transformer models substantially outperform lexical baselines in idiom detection, while hybrid and embedding-based approaches are effective for retrieval tasks.

Contributions

This paper makes the following contributions:

- We introduce IdiomX, a bilingual English–Arabic dataset containing 174,956 contextualized examples across 12,823 idioms.
- We propose a reproducible and scalable construction pipeline that combines curated resources, controlled LLM-based enrichment, and structured validation.
- We provide rich semantic and linguistic annotations, including idiomatic vs. literal usage and aligned English–Arabic meanings.
- We define a unified benchmark covering three tasks: idiom detection, context-to-idiom retrieval, and cross-lingual Arabic-to-English idiom retrieval.
- We present strong baselines and detailed analyses, showing that contextual modeling, hybrid retrieval, and task-specific fine-tuning are essential for robust idiom understanding.

II. Related Work

Research on idiomatic language understanding has evolved along two main directions:

(1) the construction of annotated datasets and

(2) the development of computational models for idiom detection and interpretation.

A. *Idiom Datasets*

Early work on idiom understanding focused on relatively small-scale annotated datasets. The VNC-Tokens dataset [1] and the SemEval-2013 Task 5b dataset [2] established standard benchmarks for distinguishing between idiomatic and literal usage. However, these datasets are limited in scale and contextual diversity, which restricts their ability to support modern data-driven models.

Subsequent efforts expanded both coverage and dataset size. The PIE Corpus [3] introduced a larger collection of idiomatic expressions with contextual annotations, while MAGPIE [4] further increased scale and diversity. Despite these improvements, most existing datasets remain predominantly monolingual and are typically designed for a single task, most commonly idiom detection.

In addition, many datasets focus on specific idiom types, such as verb–noun constructions (e.g., “take place”), fixed expressions (e.g., “by and large”), or semi-compositional phrases (e.g., “spill the beans”). This limits their ability to generalize across broader categories of figurative language.

Recent work has explored multilingual and cross-lingual idiom resources [5], [6]. While these approaches aim to capture semantic equivalence across languages, they often rely on direct translation and lack large-scale contextual examples, balanced idiomatic versus literal supervision, and explicit modeling of ambiguity.

Overall, there remains a clear need for a large-scale, context-rich, bilingual dataset that supports multiple tasks within a unified evaluation framework—this gap motivates the development of IdiomX

B. *Models for Idiom Understanding*

A wide range of computational approaches have been proposed for idiom understanding. Early methods relied on surface-level lexical features and statistical models, such as n-gram representations combined with traditional classifiers. While simple and efficient, these approaches are limited in their ability to capture contextual meaning.

More recent work has focused on neural architectures, particularly transformer-based models such as BERT [13]. These models use self-attention mechanisms to capture contextual relationships between words, enabling them to better distinguish between literal and idiomatic usage.

In multilingual settings, cross-lingual embedding models and multilingual transformers have enabled semantic alignment across languages. Approaches based on Sentence-BERT [17] and multilingual representation learning [6] have shown strong performance in semantic similarity and retrieval tasks.

However, the effectiveness of these models is highly dependent on the availability of large-scale, high-quality datasets. Without sufficient contextual diversity and balanced supervision, even advanced models struggle with ambiguity and generalization in idiomatic language

C. *Limitations of Existing Work*

Despite substantial progress, several key limitations remain in existing idiom research:

(1) limited dataset scale,
(2) insufficient contextual diversity,
(3) lack of bilingual or cross-lingual alignment, and
(4) narrow task coverage, often focused on a single problem such as idiom detection.

These limitations restrict both the development and evaluation of models, particularly in scenarios that require contextual reasoning or cross-lingual understanding.

D. IdiomX in Context

To address these challenges, we introduce IdiomX, a large-scale bilingual idiom benchmark that integrates multiple desirable properties within a single dataset. IdiomX provides aligned English–Arabic semantic representations, includes idiomatic and literal usage examples, and supports multiple tasks including idiom detection, context-to-idiom retrieval, and cross-lingual retrieval.

Unlike prior datasets such as PIE [3] and MAGPIE [4], IdiomX combines large-scale contextual coverage with controlled generation, semantic validation, and minimal duplication (reuse factor ≈ 1.04). In addition, IdiomX captures a wide range of contextual variations for each idiom, enabling more realistic and robust evaluation.

These properties position IdiomX as a comprehensive benchmark for studying idiomatic language understanding in both monolingual and multilingual settings.

Table 1: Comparison of IdiomX with widely used idiom datasets in terms of scale, multilingual support, and task coverage.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	N	ACL Anthology
SemEval-2013 Task 5b [10]	EN	85	~4,350	N	ACL Anthology
IDIX [9]	EN	52	4,022	N	ACL Anthology
MAGPIE [8]	EN	~1,700	56,000	N	ACL Anthology
PIE-English [11]	EN	~1,200	~20,100	N	arXiv
IdiomX [23]	EN+AR	12,823	174,956	Y	Public Repository [1]

Table 1: Comparison of IdiomX with existing idiom datasets in terms of scale, multilingual support, and task coverage.

As shown in Table 1, IdiomX significantly surpasses existing datasets in both scale and coverage, providing 174,956 contextualized examples across 12,823 idioms. It is the only dataset that simultaneously combines:

• large-scale contextual diversity
• bilingual semantic alignment (English–Arabic)
• support for multiple benchmark tasks within a unified framework

In contrast to prior resources, which are typically smaller, monolingual, or designed for a single task, IdiomX enables comprehensive evaluation of idiomatic language understanding across multiple dimensions

This design positions IdiomX as a next-generation benchmark for idiom detection, semantic retrieval, and cross-lingual language understanding.

III. DATASET CONSTRUCTION

The IdiomX dataset is constructed through a multi-stage pipeline that combines curated linguistic resources with controlled large language model (LLM) enrichment and structured validation.

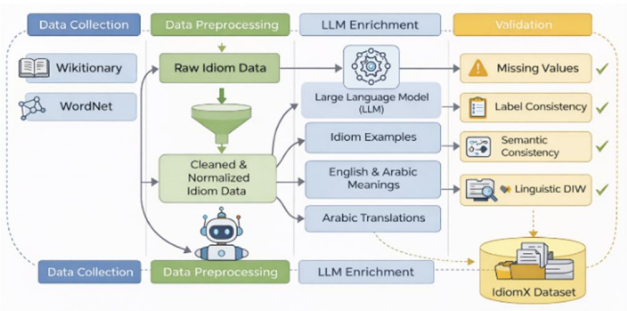

Figure 1 illustrates the overall dataset construction pipeline.

A. Data Sources and Collection

The construction process begins by aggregating idiomatic expressions from multiple linguistic resources, including publicly available idiom dictionaries and prior datasets such as VNC-Tokens [1], SemEval-2013 Task 5b [2], PIE [3], and MAGPIE [4]. This step aims to maximize coverage across different types of idiomatic expressions, including verb–noun constructions, fixed expressions, and semi-compositional phrases.

Each collected expression is normalized into a canonical form to ensure consistency. Surface variations and duplicates are removed, and each idiom is assigned a unique identifier (idiom_id) along with its canonical representation (idiom_canonical). This standardized representation is used throughout the dataset.

B. LLM-based Dataset Enrichment

To expand coverage and improve contextual diversity, we perform large-scale dataset enrichment using the OpenAI GPT-4.1-mini model [25], a lightweight variant designed for efficient high-volume generation.

The enrichment process is implemented as a batch-based asynchronous pipeline. Each idiom is converted into a structured prompt and processed at scale, enabling efficient and reproducible data generation.

For each idiom, the model generated:

- Natural example sentences
- Arabic translations of the examples
- English and Arabic meanings
- Bilingual explanations
- Usage labels (idiomatic, literal, borderline)
- Adversarial or challenging examples

This process significantly increases dataset size while preserving linguistic quality and semantic consistency

C. Prompt Design

We design a structured prompt to ensure consistency, realism, and semantic correctness across all generated examples.

Key constraints included:

- The idiom must appear naturally within the sentence.
- Sentences must be grammatically correct and contextually realistic.
- Arabic translations must be fluent and semantically accurate.
- Meanings must reflect the intended idiomatic interpretation.
- Usage labels must match the actual usage in context.

In addition, challenging examples are introduced to create harder evaluation cases, improving robustness for downstream tasks.

This approach follows best practices in controlled LLM generation and instruction-based prompting [25], ensuring both scalability and data quality.

D. Automated Validation and Correction

Following generation, all examples undergo a second-stage validation process using the same LLM in a structured verification mode.

Each generated instance is evaluated against strict criteria:

- Linguistic correctness (English and Arabic)
- Consistency between the idiom and the example sentence
- Alignment between usage labels and intended meaning
- Accuracy of explanations
- Structural completeness of all fields

[1] [1] Dataset and code available at:
https://huggingface.co/datasets/aymansharara/IdiomX
https://github.com/aymanshar/idiomx-dataset

Validation outputs are enforced using a predefined JSON schema, ensuring consistent and machine-readable results [25].

Each entry is assigned one of the following statuses:

- **verified**: accepted without modification.
- **corrected**: automatically refined to resolve issues.
- **invalid**: rejected due to major linguistic or semantic errors,

The validation pipeline supports checkpointing and incremental processing, enabling reliable and scalable handling of large datasets.

E. Quality Control Strategy

Quality control is enforced through multiple complementary layers:

1. **Pre-filtering**
   - Removal of non-idiomatic expressions.
   - Deduplication of examples.
2. **Controlled LLM Generation**
   - Strict prompt design and constraints.
   - Schema-constrained structured outputs.
3. **Post-generation Validation**
   - Automated correction of inconsistent entries.
   - Detection of semantic or grammatical errors.
4. **Statistical Validation**
   - Analysis of label distribution, sentence length, and coverage.
   - Ensures dataset consistency and balance [26].

F. Examples of Dataset Quality

Idiomatic example
"He finally kicked the bucket after a long illness."
Meaning: To die.

Literal example
"She kicked the bucket across the yard."

Low-quality example (rejected or corrected)
"He kicked bucket yesterday very sadness."
This example is rejected due to grammatical and semantic errors.

G. Final Dataset Characteristics

The final version of IdiomX contains 174,956 contextualized examples covering 12,823 English idioms, with 172,393 unique sentences and a reuse factor of approximately 1.04, indicating minimal duplication.

Each idiom is represented by an average of approximately 14 contextual examples, ensuring strong contextual diversity.

The dataset includes a balanced distribution of idiomatic and literal usage, supported by aligned English–Arabic semantic annotations, including meanings, explanations, and usage labels.

These characteristics make IdiomX suitable for multiple tasks, including idiom detection, context-to-idiom prediction, and cross-lingual Arabic-to-English idiom retrieval.

IV. DATASET ANALYSIS

In this section, we present a comprehensive analysis of the IdiomX dataset, examining its linguistic properties, quality, and distributional characteristics. The analysis highlights the dataset's suitability for robust idiom understanding and cross-lingual modeling.

A. **Compositionality Distribution**

Compositionality measures the extent to which the meaning of an idiom can be inferred from its constituent words. As shown in Fig. 2, the dataset is dominated by semi-opaque idioms (54.63%), followed by opaque expressions (33.38%) and transparent ones (11.99%). This distribution reflects realistic language usage, where most idioms exhibit partial compositionality rather than being fully opaque or fully transparent.

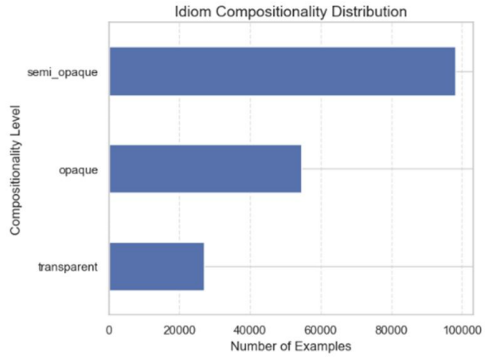

Figure 2 compositionality distribution.

B. **Register Distribution**

Register indicates the formality level of expressions. As shown in Fig. 3, neutral register dominates the dataset (59.9%), followed by informal usage (34.7%) and formal expressions (5.4%). This distribution ensures strong coverage of everyday conversational language while still including formal contexts, making the dataset applicable across diverse real-world scenarios.

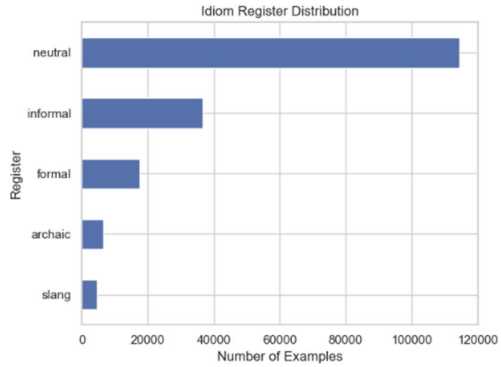

Figure 3 idiom register distribution.

C. **Difficulty Distribution**

Difficulty levels capture the learning complexity of idioms. As shown in Fig. 4, most idioms fall within intermediate and advanced levels, with fewer beginner-level expressions. This distribution reflects the inherently challenging nature of idiomatic language and ensures that the dataset supports realistic evaluation of model capabilities.

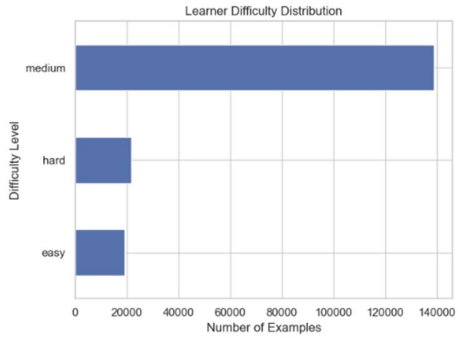

Figure 4 idiom learner difficulty distribution.

D. Semantic Quality Distribution

Semantic quality measures the correctness and coherence of generated examples. As shown in Fig. 5, over 77% of examples are rated as high quality, with a smaller proportion classified as medium or low quality. This confirms the effectiveness of the LLM-based generation and validation pipeline in producing reliable and meaningful data.

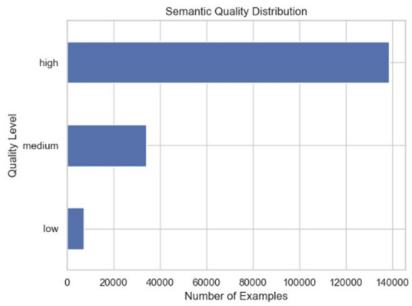

Figure 5 semantic quality distribution.

E. Sentence Length Distribution

Sentence length affects contextual complexity and model processing difficulty. As shown in Fig. 6, the dataset exhibits a well-balanced distribution of sentence lengths, covering both short and long contexts. This variability enhances the dataset's ability to support models across different levels of contextual reasoning.

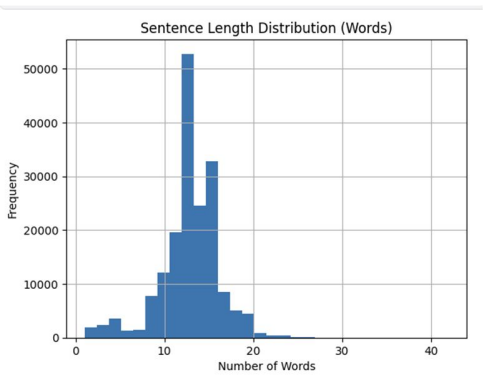

Figure 6 sentence length distribution.

F. Semantic Similarity Distribution

Semantic similarity captures how closely generated examples relate to the canonical idiom meaning. As shown in Fig. 7, the dataset includes a wide range of similarity levels, including high-similarity examples and more challenging borderline cases. This diversity is essential for training models capable of distinguishing subtle semantic differences.

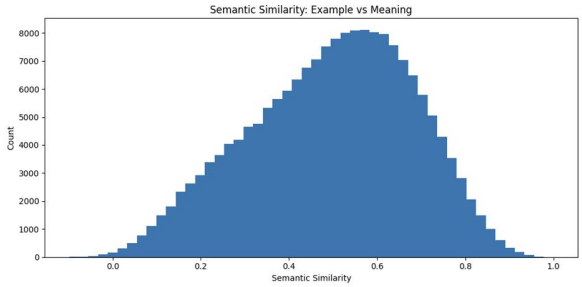

Figure 7 semantic similarity analysis.

G. *Idiomatic vs Literal Balance*

IdiomX maintains a balanced distribution between idiomatic and literal usage, ensuring that models are not biased toward a single interpretation. This balance is critical for classification and disambiguation tasks, where distinguishing between literal and figurative meaning is the primary objective.

H. *Contextual Diversity*

Each idiom is represented across multiple discourse types, including dialogue, narrative, formal, social media, question, and sarcastic contexts. This controlled variation ensures that models learn idioms in diverse linguistic environments rather than memorizing fixed patterns.

I. *Cross-Lingual Alignment*

All examples include aligned English and Arabic meanings and explanations, enabling cross-lingual evaluation. This bilingual structure supports research in multilingual representation learning and semantic transfer between languages.

J. *Key Dataset Strengths*

To summarize, IdiomX provides several key advantages over existing datasets:

- Large-scale coverage with 174,956 contextualized examples
- High contextual diversity with minimal duplication (reuse factor ≈ 1.04)
- Balanced idiomatic and literal usage
- High semantic quality with over 77% high-quality samples
- Structured generation ensuring consistency
- English–Arabic alignment for cross-lingual evaluation
- Inclusion of borderline cases for modeling ambiguity

These properties make IdiomX particularly suitable for modern NLP tasks requiring robustness, generalization, and multilingual understanding.

V. Benchmark Tasks

To comprehensively evaluate idiom understanding, we define three benchmark tasks that capture complementary aspects of figurative language processing, including classification, contextual reasoning, and cross-lingual semantic alignment.

Together, these tasks move beyond isolated evaluation settings and provide a unified framework for studying idiomatic language in both monolingual and multilingual scenarios.

A. Task 1: Idiom Detection

Idiom detection focuses on identifying whether an expression in a sentence is used idiomatically or literally, which is a fundamental challenge in figurative language understanding [1], [2].

The task is formulated as a binary classification problem. Each input consists of a sentence containing a target expression, and the model predicts whether the usage is idiomatic **or** literal.

This task evaluates a model's ability to distinguish figurative meaning from surface-level interpretation, particularly in cases where the same expression can have both literal and idiomatic meanings depending on context.

B. Task 2: Context-to-Idiom Retrieval

Given a contextual sentence, the goal is to retrieve the most appropriate idiom that best matches the underlying meaning.

We formulate this as a constrained retrieval problem over a predefined idiom inventory. Models are required to map contextual semantics to the correct idiomatic expression.

This task evaluates deeper semantic understanding and expressive alignment, requiring models to capture intent beyond literal wording and identify the most suitable figurative expression.

C. Task 3: Arabic-to-English Idiom Retrieval

This task extends retrieval to a cross-lingual setting, where the input is an Arabic sentence and the goal is to retrieve the corresponding English idiom.

It is formulated as a ranking problem, where candidate idioms are scored based on their semantic similarity to the input.

This task evaluates a model's ability to bridge meaning across languages, particularly for figurative expressions that often lack direct lexical correspondence [6].

Overall, these tasks form a progressive evaluation framework, moving from monolingual classification (Task 1) to contextual semantic retrieval (Task 2), and finally to cross-lingual semantic alignment (Task 3).

VI. EXPERIMENTS AND RESULTS

This section presents the experimental evaluation of the IdiomX dataset across the defined benchmark tasks. The goal is to assess how well the dataset supports different problem types, including idiom detection, context-to-idiom retrieval, and cross-lingual retrieval.

We evaluate both classification and retrieval performance using standard metrics widely used in natural language processing. The results provide insights into how different modeling approaches handle idiomatic language, which often requires deeper contextual understanding beyond surface-level text.

A. Experimental Setup

The IdiomX dataset is split into training, validation, and test sets using a stratified approach to ensure balanced distribution of idiomatic and literal examples across all splits.

Each idiom appears in both idiomatic and literal contexts, creating a controlled and fair evaluation setting.

For classification tasks, we report:

- Accuracy
- Precision, Recall, and F1-score

For retrieval tasks, we use:

- Top-1 accuracy
- Top-K accuracy

These metrics allow evaluation of both exact predictions and ranking quality.

We evaluate three types of models:

- **TF-IDF + Logistic Regression (lexical baseline) -** captures surface-level patterns without contextual understanding.
- **DistilBERT [12] -** a lightweight transformer model used as a strong baseline.
- **RoBERTa [14] -** a larger transformer model designed to capture deeper contextual relationships.

For cross-lingual tasks, multilingual models such as XLM-R [6] are used, as they are designed to handle multiple languages effectively.

All experiments are conducted using the same preprocessing pipeline and consistent training settings to ensure a fair comparison between models.

We begin with Task 1 (Idiom Detection), which serves as the primary classification benchmark for evaluating contextual understanding

B. Task 1: Idiom Detection

1. Problem Definition

The goal of this task is to determine whether a sentence contains an idiomatic or a literal usage of an expression. This is challenging because idiomatic meaning cannot be directly inferred from individual words and depends heavily on context [1], [2].

We formulate this as a binary classification problem, where each sentence is labeled as either *idiomatic* or *literal*.

2. Models

We evaluate three models with increasing levels of contextual understanding:

• **TF-IDF + Logistic Regression (Baseline)**

A traditional machine learning approach that represents text using word frequency features. This model captures surface-level lexical patterns but does not model context.

• **DistilBERT [12] (Transformer Baseline)**

A compact transformer model that uses self-attention to capture contextual relationships between words. It provides a strong balance between performance and efficiency.

• **RoBERTa [14] (Improved Model)**

A larger and more powerful transformer model trained with optimized pretraining strategies. It is designed to capture deeper contextual and semantic relationships in text.

All models are trained and evaluated under the same conditions to ensure a fair and consistent comparison.

3. *Data Processing*

Input sentences were tokenized using the pretrained tokenizer corresponding to each model. The following preprocessing settings were applied:

- Padding to a fixed maximum length of 128 tokens
- Truncation of longer sequences

The dataset was split using a stratified approach to ensure balanced distribution of idiomatic and literal examples. The Hugging Face *datasets* library was used for efficient preprocessing and data handling.

label	Count	Percentage (%)
literal	84,374	50.74
idiomatic	81,905	49.26

Table 2: Class Distribution of Idiomatic vs Literal Examples.

The dataset is well balanced, ensuring that evaluation is not biased toward a dominant class.

4. *Training Setup*

All models were fine-tuned using the same configuration:

- Batch size: 8
- Learning rate: 2e-5
- Number of epochs: 3
- Evaluation performed at the end of each epoch
- Best model selected based on validation F1-score

This setup ensures stable training and enables a fair comparison between models.

5. *Results*

A. *Model Comparison*

Model	Accuracy	F1-score
TF-IDF + Logistic egression	0.8732	0.8708
DistilBERT	0.9227	0.9213
RoBERTa	0.9306	0.9294

Table 3: Model Performance on Idiom Detection Task

Transformer-based models significantly outperform the lexical baseline, demonstrating the importance of contextual understanding for idiom detection.

RoBERTa achieves the best performance, with an accuracy of 93.06% and F1-score of 0.929, confirming the benefit of larger and more expressive models.

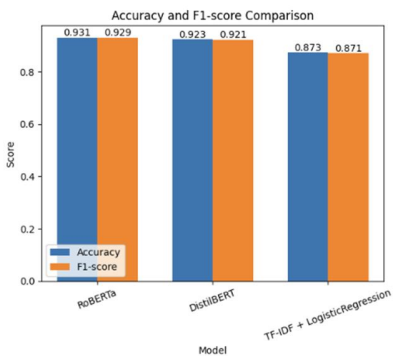

Figure 8: Model Accuracy and F1-score comparison on idiom detection

DistilBERT provides a strong trade-off between performance and efficiency, achieving competitive results despite its smaller size

B. *Detailed Performance Analysis*

RoBERTa shows consistent improvement over DistilBERT across all metrics. In particular:

• higher precision → fewer false positives
• balanced recall → stable detection of idiomatic usage
• improved F1-score → better overall classification quality

This indicates that deeper contextual representations are critical for distinguishing subtle differences between literal and idiomatic meaning.

6. *Error Analysis and Key Failure Modes*

To better understand model limitations, we analyze prediction errors using both quantitative and qualitative approaches. In addition to examining misclassified examples, we analyze confidence scores and error distributions to identify systematic patterns:

(1) Limited or Insufficient Context

Short or context-poor sentences make it difficult to determine whether an expression is used idiomatically or literally. Without sufficient surrounding information, the model may rely on surface patterns rather than meaning.

(2) Strong Idiom Bias

Highly frequent idioms (e.g., “kick the bucket”) are sometimes predicted as idiomatic even when used literally. This indicates that the model partially memorizes common idioms and overgeneralizes them across contexts.

This behavior is also reflected in the confusion matrix, where false positives are slightly more frequent than false negatives

(3) Context-Dependent Ambiguity

Some expressions can be interpreted as either literal or idiomatic depending on subtle contextual cues. These borderline cases are particularly challenging, as they require fine-grained semantic understanding beyond explicit wording.

(4) Indirect or Implied Meaning

In certain cases, idiomatic meaning is not explicitly stated but implied. The model may fail to capture such implicit meanings, leading to false negatives.

(5) Overconfidence in Incorrect Predictions

Analysis of prediction confidence reveals that the model often assigns high confidence scores even to incorrect predictions.

The mean confidence for incorrect predictions is approximately **0.908**, while correct predictions reach approximately **0.988**, indicating strong confidence separation but also highlighting cases of overconfidence.

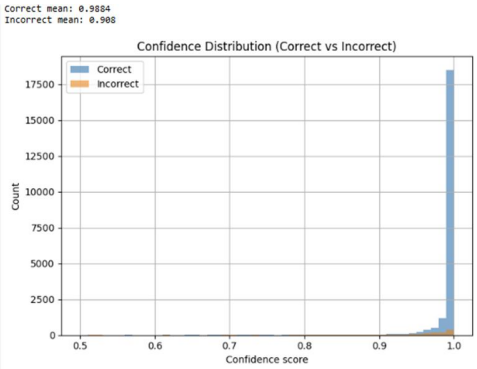

Figure 9: Confidence distribution for correct vs incorrect predictions.

As shown in Figure 9, most correct predictions are concentrated near high confidence values, while incorrect predictions, although fewer, still appear at high confidence levels.

(6) Error Distribution by Confidence

To further analyze model reliability, we examine how errors are distributed across confidence scores.

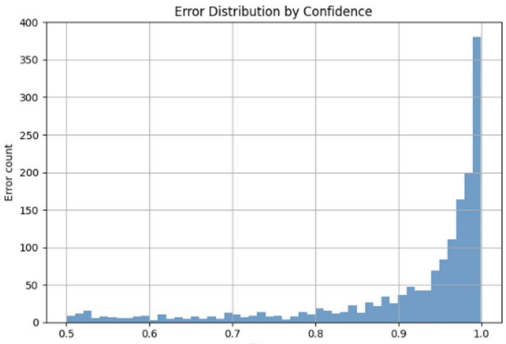

Figure 10: Error distribution by confidence.

The majority of errors occur at high confidence levels, indicating that the model is often confidently wrong in difficult or ambiguous cases. This suggests that confidence alone is not always a reliable indicator of correctness.

(7) Hard Negative Analysis

We analyze the most confident incorrect predictions (hard negatives) to better understand challenging cases.

	example	y_true	y_pred	confidence
75851	The scientist stuck their nose into the micros...	1	0	0.999517
62576	I pressed the fabric into service as a makeshi...	1	0	0.999515
43134	OMG they are always together! Literally joined...	1	0	0.999505
16230	How do pioneers circle the wagons to protect t...	0	1	0.999493
18487	Yeah, he literally couldn't get elected dogcat...	1	0	0.999493
12506	Is that rusty old bucket of bolts still runnin...	1	0	0.999470
61693	Yeah, they went behind the barn to plough the ...	1	0	0.999464
73627	The equipment was soaking wet due to the unexp...	1	0	0.999460
108925	Did you accidentally pocket dial your friend w...	1	0	0.999455
109612	Could you put the wood in the hole before you ...	1	0	0.999452

Figure 11: Examples of high-confidence errors.

Many hard negatives involve sentences where:
• both literal and idiomatic interpretations are plausible
• subtle contextual cues determine the correct meaning
• the model fails to fully capture semantic nuance

In several cases, prediction confidence exceeds **0.99**, indicating strong overconfidence in ambiguous contexts.

Overall, these findings show that while transformer models capture contextual information effectively, they still struggle with ambiguity, implicit meaning, and fine-grained semantic distinctions.

These observations are consistent with prior work on figurative language understanding [3], which highlights the difficulty of modeling non-compositional meaning.

7. Qualitative Examples

To further illustrate model behavior, we present representative examples covering both correct predictions and common failure cases:

Example 1 (Correct – Idiomatic)
"He finally kicked the bucket after a long illness."
Prediction: Idiomatic

Analysis:

The model correctly identifies the figurative meaning based on strong contextual cues indicating death. This demonstrates effective contextual understanding for well-formed idiomatic usage.

Example 2 (Correct – Literal)
"She kicked the bucket across the yard."
Prediction: Literal

Analysis:

The sentence clearly describes a physical action, and the model correctly distinguishes it from idiomatic usage. This indicates that the model can recognize explicit literal contexts.

Example 3 (Error – False Positive)
"He kicked the bucket lightly to move it aside."
Prediction: Idiomatic
Actual: Literal

Analysis:

The model incorrectly predicts idiomatic usage due to the presence of a highly frequent idiom. This reflects a bias toward memorized patterns, where the model over-relies on the expression itself rather than fully interpreting the surrounding context.

Example 4 (Error – False Negative)
"After years of stress, he finally let everything go."
Prediction: Literal
Actual: Idiomatic

Analysis:

The idiomatic meaning is implied rather than explicitly stated, making it more difficult to detect. This highlights a limitation in capturing subtle or indirect figurative expressions.

These examples confirm that most errors arise from contextual ambiguity and implicit meaning rather than random misclassification.

In particular:

- false positives are often driven by idiom memorization bias
- false negatives are linked to implicit or weak contextual signals

These observations are consistent with the error patterns discussed in the previous section and further emphasize the challenge of modeling non-compositional language.

8. Confusion Matrix Analysis

The confusion matrix provides a detailed view of model performance across idiomatic and literal classes.

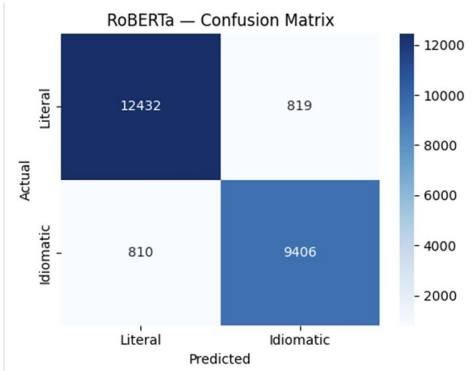

Figure 12: Confusion matrix for RoBERTa on the test set.

As shown in Figure 12, the model achieves strong performance for both classes, with high true positive rates for both idiomatic and literal examples.

However, a slightly higher number of false positives can be observed (literal → idiomatic). This indicates a mild tendency of the model to classify expressions as idiomatic, even when they are used literally.

This behavior is consistent with the previously identified idiom frequency bias, where well-known expressions are more likely to be interpreted as idiomatic.

Overall, the confusion matrix confirms balanced performance, with a small but consistent bias toward idiomatic predictions.

9. ROC Curve Analysis

To further evaluate classification performance across different decision thresholds, we analyze the Receiver Operating Characteristic (ROC) curve.

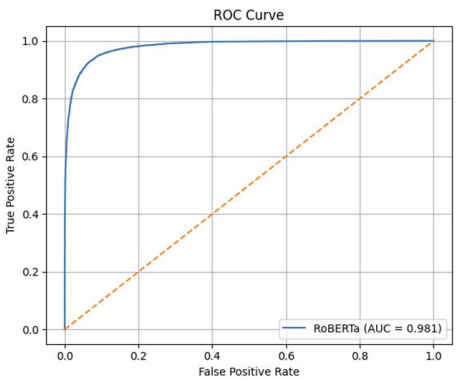

Figure 13: ROC curve for RoBERTa (AUC = 0.981).

The ROC curve shows a strong separation between classes, with an Area Under the Curve (AUC) of **0.981**, indicating excellent discriminative ability.

The curve remains close to the top-left corner, demonstrating that the model achieves a high true positive rate while maintaining a low false positive rate across thresholds.

This result confirms that the model is robust and reliable in distinguishing between idiomatic and literal usage, even when decision thresholds vary.

10. Ablation Study

To understand the impact of model capacity, we compare the performance of DistilBERT and RoBERTa.

RoBERTa achieves higher accuracy and F1-score, indicating stronger contextual understanding and better ability to capture subtle semantic differences.

In contrast, DistilBERT achieves slightly higher recall, suggesting that it is more permissive and tends to classify more examples as idiomatic, leading to increased false positives.

This trade-off reflects a common pattern in transformer models:

• smaller models generalize more aggressively
• larger models produce more precise and balanced predictions

These results confirm that increased model capacity improves idiom detection performance, particularly for capturing nuanced contextual cues.

11. Conclusion

The results demonstrate that transformer-based models are highly effective for idiom detection, where contextual understanding is essential. While DistilBERT provides a strong and efficient baseline, RoBERTa consistently achieves the best performance across all key metrics, including accuracy and F1-score.

These findings show that idiomatic language cannot be reliably modeled using surface-level patterns alone, and instead requires deeper contextual and semantic representations. Larger transformer architectures are better suited to capture these nuances, leading to more accurate and balanced predictions.

In addition, the analysis of confidence scores and hard negative examples highlights that ambiguity and implicit meaning remain key challenges, even for high-performing models

Overall, the results confirm that IdiomX provides a robust and reliable benchmark for idiom detection. The dataset enables comprehensive evaluation of contextual language understanding and supports the development of more advanced models for figurative language processing.

C. Task 2: Context-to-Idiom Retrieval

This task evaluates whether a model can recover the correct canonical idiom from an English contextual sentence that uses the idiom idiomatically. Unlike Task 1, which is a binary classification problem, Task 2 is formulated as a closed-set retrieval problem over a fixed idiom inventory.

Formally, given a contextual query x, the goal is to retrieve the correct idiom y* from a fixed idiom inventory I:

$$y^* = \text{argmax}_{y \in I} \text{score}(x, y)$$

where (s(x, i)) is the retrieval score assigned to idiom (i) given context (x).

This formulation requires both semantic understanding of the contextual sentence and precise matching to a canonical idiom form. Because the candidate inventory is fixed, the task provides a controlled and reproducible benchmark for evaluating retrieval-based idiom understanding.

1. Benchmark Construction and Split Strategy

The Task 2 benchmark is constructed from the IdiomX high-quality subset. To ensure a reliable retrieval setting, we keep only examples that satisfy the following conditions:

(1) the usage label is idiomatic,
(2) validation_status = valid, and
(3) semantic_quality = high.

After filtering, the final benchmark contains 38,604 examples associated with an idiom bank of 11,626 canonical idioms.

To prevent sentence leakage between training and evaluation, we apply a query-level split. Under this strategy, the same contextual sentence is never shared across splits. However, the same idiom may appear in both training and test sets through different contextual realizations, which is appropriate for the closed-set retrieval setting because the target inventory is fixed at evaluation time.

The final split contains 30,883 training examples and 7,721 test examples, with 5,356 idioms shared between train and test. This design ensures evaluation on unseen contexts while preserving realistic idiom coverage across both partitions.

Setting	Value
Dataset variant	IdiomX high-quality
Filtering	idiomatic only; valid; high semantic quality
Total examples	38,604
Idiom bank size	11,626
Train examples	30,883
Test examples	7,721
Shared idioms between train and test	5,356
Split strategy	Query-level
Sentence leakage across splits	None

Table 5: Task 2 benchmark statistics and split configuration.

2. Retrieval Architecture

We evaluate three retrieval settings of increasing complexity.

- **MiniLM dense retrieval**.

In the first setting, both contextual sentences and candidate idioms are encoded into a shared embedding space using MiniLM sentence embeddings [28]. Retrieval is then performed by ranking idioms according to semantic similarity between the input context and each idiom representation.

- **Hybrid retrieval.**

In the second setting, dense retrieval is combined with BM25 [27] sparse retrieval. The dense component captures semantic similarity beyond surface wording, while the sparse component captures direct lexical overlap between the context and idiom expressions. Their combination improves candidate recall by leveraging both semantic and lexical evidence.

- **Hybrid retrieval with reranking.**

In the final setting, a cross-encoder reranker [29] is applied to the top-10 candidates returned by the hybrid retriever. The reranker jointly scores each context–idiom pair, allowing more precise discrimination among semantically similar candidates. This produces the final ranked list used for evaluation.

This multi-stage design follows a standard retrieval pipeline in which early retrieval stages focus on candidate recall, while the final reranking [29] stage improves ranking precision.

3. Evaluation Metrics

Performance is evaluated using standard ranking metrics: Top-1 accuracy**,** Top-3 accuracy**,** Top-5 accuracy**,** and Mean Reciprocal Rank (MRR)**.**

Top-k accuracy measures whether the gold idiom y^* appears within the first k retrieved candidates. MRR evaluates the average reciprocal rank of the first correct prediction, providing a more sensitive measure of ranking quality.

The metrics subsection is fine as written, but if you want it to look slightly more formal, you can add one reference on ranking evaluation practice. This is optional.

4. Quantitative Results

Table 6 reports the main results for Task 2 under the closed-set retrieval setting. We compare three retrieval configurations: MiniLM dense retrieval, hybrid retrieval using MiniLM and BM25, and hybrid retrieval followed by cross-encoder reranking. MiniLM serves as the dense semantic baseline, while BM25 adds lexical matching signals that are known to remain useful in retrieval settings [17], [27]. The reranking stage further refines the top retrieved candidates using a cross-encoder scoring model, following standard multi-stage retrieval practice [29].

Model	Top-1	Top-3	Top-5	MRR
MiniLM	0.538	0.663	0.707	0.618
Hybrid (MiniLM + BM25)	0.763	0.871	0.901	0.825
Hybrid + reranker	0.867	0.922	0.926	0.894

Table 6: Main Task 2 retrieval results on the closed-set benchmark.

The results show a clear and consistent improvement across all evaluation metrics as the retrieval pipeline becomes more selective. MiniLM [28] achieves a Top-1 accuracy of 0.538 and an MRR of 0.618, indicating that dense retrieval alone provides a reasonable semantic baseline but still struggles with fine-grained ranking among semantically similar idioms. When BM25 is combined with MiniLM [28], performance increases substantially to 0.763 Top-1 and 0.825 MRR, showing that lexical matching and dense semantic similarity provide complementary evidence.

The strongest results are obtained with the full Hybrid + Reranker model, which reaches 0.867 Top-1, 0.922 Top-3, 0.926 Top-5, and 0.894 MRR. This confirms that the reranking stage is especially effective for resolving ambiguity among the strongest initial candidates. In other words, the hybrid retriever improves candidate recall, while the cross-encoder improves final ranking precision [29].

Figure 12 visualizes the overall comparison across Top-1, Top-3, Top-5, and MRR. The same trend is shown more clearly in Figure 13, where the Top-K retrieval curve demonstrates that each additional retrieval stage improves ranking quality, with

the largest gains appearing at Top-1. This is important because exact canonical idiom recovery is the main objective of the task.

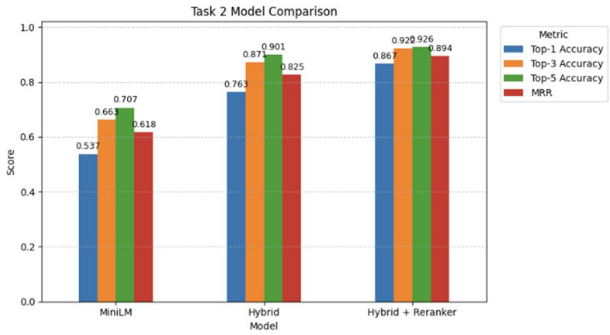

Figure 12: Performance comparison of Task 2 retrieval models across Top-1, Top-3, Top-5, and MRR.

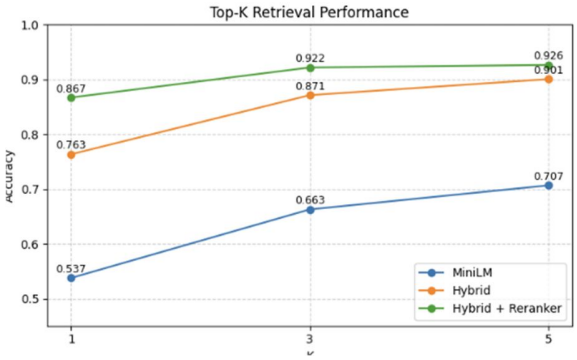

Figure 13: Top-*K* retrieval curves for Task 2 under the closed-set retrieval setting.

To quantify the improvement over the dense baseline, Table 7 and Figure 14 report the absolute gains relative to MiniLM [28]. The hybrid model improves Top-1 by **+0.226**, while the full reranking pipeline improves it by +0.329. Similar gains are observed for Top-3, Top-5, and MRR. These results indicate that context-to-idiom retrieval is not solved by semantic embedding alone; it benefits from combining semantic retrieval, lexical matching, and fine-grained pairwise reranking [17], [27], [29].

Metric	Hybrid Gain	Hybrid + Reranker
Top-1 Accuracy	+0.226	+0.329
Top-3 Accuracy	+0.209	+0.259
Top-5 Accuracy	+0.194	+0.220
MRR	+0.208	+0.276

Table 7: Absolute improvement over the MiniLM dense retrieval baseline.

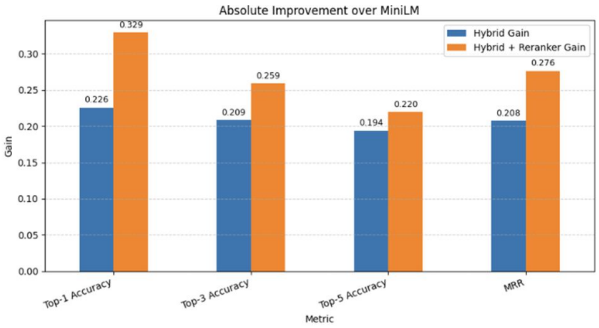

Figure 14: Absolute improvement over the MiniLM dense retrieval baseline on Task 2.

5. Error Analysis and Failure Modes

The best-performing model (Hybrid + Reranker) achieves strong performance, with 0.867 Top-1 accuracy and 0.894 MRR. However, a detailed analysis of the test set reveals 1,027 incorrect predictions out of 7,721 samples, as summarized in Figure 15. This corresponds to approximately 13.3% residual error, indicating that the task remains challenging despite high overall accuracy.

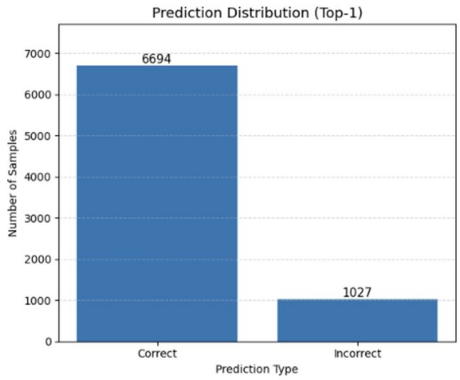

Figure 15: Distribution of correct and incorrect predictions for Task 2 on the test set.

A qualitative inspection of the errors reveals several consistent patterns.

First, the model often confuses idioms that are lexically or semantically similar, selecting plausible but incorrect expressions. Second, there is a noticeable frequency bias, where common idioms are preferred over less frequent but correct ones, a known effect in retrieval systems [27]. Third, the model sometimes relies on surface-level keyword matching, especially when lexical overlap is strong but semantically misleading. Finally, performance degrades on short or ambiguous contexts, where insufficient information is available to disambiguate between competing idioms.

Overall, these errors suggest that while the retrieval pipeline effectively combines semantic and lexical signals, idiom understanding still requires fine-grained contextual reasoning and improved handling of long-tail expressions [17], [29].

6. Qualitative Analysis

To better understand model behavior, we present qualitative examples from the test set illustrating both successful retrievals and typical failure cases. Table 8 shows representative examples produced by the best-performing model (Hybrid + Reranker).

The examples highlight that the model is generally able to recover the correct idiom when the contextual signal is strong and unambiguous. In such cases, the correct idiom is ranked at the top position, demonstrating effective semantic matching between the input context and the idiom meaning.

However, failure cases reveal consistent challenges. In particular, the model tends to select idioms that are lexically similar but semantically incorrect, or more frequent expressions that partially match the context. These errors confirm the patterns identified in Section 7, where lexical overlap and frequency bias can override deeper semantic understanding.

Overall, the qualitative results show that the retrieval pipeline performs robustly in clear contexts but struggles in cases requiring fine-grained semantic discrimination or involving less frequent idioms.

Table 8 presents representative examples from the best-performing model. The first two cases illustrate typical failure modes, while the final example shows a correct prediction under clear contextual conditions.

Query (shortened)	True Idiom	Top-1	Top-3 (shortened)	✓
"…eat them for breakfast."	eat someone for breakfast	don't worry	don't worry; eat someone…; take one…	✗
"Bone apple tea… bon appétit"	bon appétit	throw a bone to	throw a bone…; wise apple; contrary…	✗
"…finally broke the ice…"	break the ice	break the ice	break the ice; ease tension; start talking	✓

Table 8: Representative qualitative examples from Task 2

7. Task 2 Summary

In this task, we introduced a closed-set context-to-idiom retrieval benchmark built on the IdiomX high-quality dataset. The task requires mapping an idiomatic contextual sentence to its corresponding canonical idiom from a fixed inventory, combining semantic understanding with precise lexical grounding.

The results show that retrieval performance improves consistently across the proposed pipeline. Dense retrieval using MiniLM [28] provides a strong semantic baseline, while hybrid retrieval significantly improves performance by incorporating lexical matching signals [17], [27]. The addition of a cross-encoder reranker yields the best results, achieving 0.867 Top-1 accuracy and 0.894 MRR, demonstrating the importance of fine-grained pairwise scoring for resolving ambiguity among candidate idioms [29].

Overall, Task 2 highlights that idiom understanding can be effectively modeled as a retrieval problem, but requires a combination of semantic representations, lexical matching, and reranking to achieve high accuracy.

8. Discussion

The results of Task 2 provide several important insights into idiom understanding as a retrieval problem.

First, while dense semantic embeddings are effective at capturing contextual meaning, they are not sufficient on their own for precise idiom recovery. The strong improvement from hybrid retrieval confirms that lexical signals remain important, even in semantically driven tasks [17], [27]. This is particularly relevant for idioms, where surface forms often carry distinctive cues.

Second, the reranking stage plays a critical role in refining predictions. By jointly modeling the interaction between the context and candidate idioms, the cross-encoder improves ranking precision and resolves ambiguities among top candidates [29]. This demonstrates the effectiveness of multi-stage retrieval architectures for complex language understanding tasks.

Third, the error analysis reveals that remaining challenges are primarily due to semantic ambiguity, frequency bias, and limited contextual information. These findings suggest that future improvements should focus on stronger contextual encoding, better handling of long-tail idioms, and training strategies that reduce reliance on frequency-based patterns.

Finally, the proposed benchmark provides a controlled and reproducible evaluation setting, enabling consistent comparison across retrieval architectures. As such, it can serve as a foundation for future research on idiom understanding, retrieval-based semantic modeling, and multilingual extensions.

D. Task 3: Arabic Context-to-Idiom Retrieval

This task evaluates whether a model can recover the correct canonical English idiom from an Arabic contextual sentence. Similar to Task 2, this is formulated as a closed-set retrieval problem over a fixed idiom inventory. However, unlike Task 2, this setting introduces a cross-lingual challenge, where the input is in Arabic and the target idiom is in English.

Formally, given an Arabic query x, the goal is to retrieve the correct idiom y^* from a fixed idiom inventory I:

$$y^* = \arg\max_{i \in I} sim(x, i)$$

where $\mathrm{sim}(x, i)$ is the similarity score between the Arabic context and the candidate idiom.

This formulation requires both semantic understanding of the Arabic context and accurate alignment to a canonical English idiom. Compared to Task 2, the task is more challenging due to cross-lingual variation and differences in idiomatic expression across languages.

As in Task 2, we adopt a retrieval-based formulation because the candidate idiom inventory is fixed, enabling a controlled and reproducible evaluation setting.

1. Formulation

Each Arabic query x and candidate idiom i are mapped into vector representations using a shared encoder:

$$h_x = Encoder(x), \qquad h_i = Encoder(i)$$

Similarity is computed using cosine similarity [5]:

$$sim(x, i) = \frac{h_x \cdot h_i}{\|h_x\| \cdot \|h_i\|}$$

Prediction is defined as:

$$\hat{y} = \arg\max_{i} sim(x, i)$$

where cosine similarity [5] is used to rank candidate idioms in the shared embedding space.

2. Experimental Setup

To construct the Task 3 benchmark, we initially explored multiple approaches, including generation-based models (mT5 [19]), hybrid generation–retrieval variants, and Arabic semantic bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple fields and applied a strict split strategy to prevent overlapping at the idiom level.

While this setup was methodologically rigorous, it resulted in near-zero retrieval accuracy, indicating that the task formulation was overly constrained and not well aligned with a retrieval-based setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem. The final benchmark retains only idiomatic examples and uses

idiom_in_example_arabic as the Arabic query and idiom_canonical as the English retrieval target.

Instead of splitting by idiom identity, we apply an example-level split, allowing the same idiom to appear in both training and test sets through different contextual realizations. This setup evaluates whether the model can generalize to unseen Arabic contexts while preserving a realistic and fixed candidate inventory.

Under this revised formulation, retrieval performance improves substantially. The fine-tuned multilingual E5 model achieves strong results (Top-1 = 64.86%, Top-5 = 81.46%), demonstrating effective cross-lingual semantic alignment.

Overall, the results show that stronger pretrained multilingual embeddings alone provide only modest gains, while task-specific contrastive fine-tuning is the primary driver of performance improvement. Additional gains are obtained through richer supervision, such as multiple positive examples per query and extended training.

The final best-performing system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. *Candidate Representation*

In Task 3, the candidate idiom inventory is represented using multiple Arabic textual views to improve semantic coverage. For each idiom, we include:

- example sentences (idiomatic usage)
- example meanings
- canonical meanings

Each representation is encoded into a shared multilingual embedding space, producing a dense representation matrix over the idiom inventory.

This multi-view representation allows the model to match Arabic queries against diverse contextual signals, improving robustness in cross-lingual retrieval.

4. *Models Evaluated*

We evaluate a set of multilingual dense retrieval models of increasing capability:

- Multilingual MiniLM [28] (baseline)
- Multilingual E5 (zero-shot)
- Multilingual E5 (fine-tuned)

MiniLM [28] serves as a lightweight semantic baseline. The E5 model introduces improved multilingual representation quality, while the fine-tuned variant is trained using contrastive learning on Arabic–idiom pairs to better align cross-lingual semantics.

While all models are evaluated, detailed analysis focuses on the best-performing configuration.

5. *Quantitative Results*

Performance is evaluated using standard ranking metrics, including Top-1 and Top-5 accuracy. Top-K accuracy measures whether the correct idiom appears within the top Kretrieved candidates.

$$Top - K = \frac{1}{N}\sum 1\{y \in TopK(x)\}$$

Table 8 reports the retrieval performance of the evaluated models.

Model	Embedding Dim	Top-1	Top-5
MiniLM-L12-v2 [28]	384	0.159	0.254
E5-base (zero-shot)	768	0.191	0.310
E5-base (finetuned)	768	0.578	0.761

Table 8: Retrieval performance on Task 3.

The results show a clear and consistent improvement across all models. MiniLM [28] provides a baseline for multilingual semantic retrieval but achieves limited performance, indicating that generic embeddings are insufficient for cross-lingual idiom understanding.

The E5 model improves performance under zero-shot settings, demonstrating the benefit of instruction-aware multilingual representations. However, the improvement remains modest, suggesting that pretrained embeddings alone are not sufficient for precise idiom retrieval.

The strongest results are obtained with the fine-tuned E5 model, which achieves 0.578 Top-1 and 0.760 Top-5 accuracy. This represents a substantial improvement over both baselines and highlights the importance of task-specific contrastive fine-tuning.

Overall, the results indicate that cross-lingual idiom retrieval requires not only strong multilingual representations but also explicit supervision to align Arabic contextual meaning with canonical idiomatic expressions.

Figure 16 illustrates the performance comparison across models, showing consistent gains at both Top-1 and Top-5.

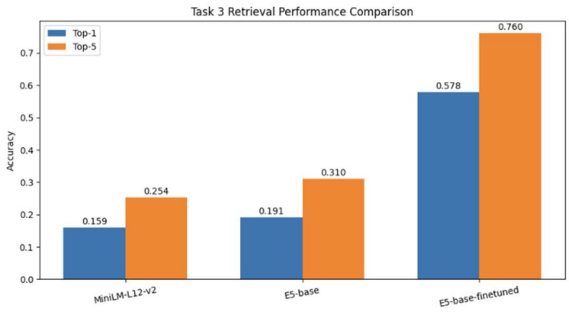

Figure 16: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. *Ablation Study and Model Progression*

The ablation study reveals a clear and consistent improvement across the retrieval pipeline, highlighting the impact of model architecture and training strategy.

As shown in Table 8, the MiniLM [28] baseline achieves limited performance, indicating that generic multilingual embeddings are insufficient for capturing idiomatic semantics in cross-lingual settings. Replacing MiniLM [28] with zero-shot multilingual E5 yields only modest gains, suggesting that pretrained multilingual representations alone are not sufficient for precise idiom alignment.

In contrast, fine-tuning the E5 model produces a substantial improvement, increasing Top-1 accuracy from 0.191 to 0.578 and Top-5 accuracy to 0.760. This represents a large relative gain over both baseline models.

These results confirm that task-specific contrastive training is the primary factor driving retrieval performance. Additional improvements are obtained through enhanced training strategies, including richer supervision and improved positive pair diversity, which contribute to better semantic alignment.

Overall, the ablation study demonstrates that idiom retrieval is not a general semantic similarity task, but a specialized problem that requires targeted supervision and fine-grained representation learning.

7. *Error Distribution Analysis*

To better understand model behavior, we categorize predictions into three outcome types:

- Top-1 correct: 6,662 queries (57.8%)
- Retrieved but misranked: 2,106 queries (18.3%)
- Fully incorrect: 2,762 queries (23.9%)

Figure 17 shows the distribution of these outcomes.

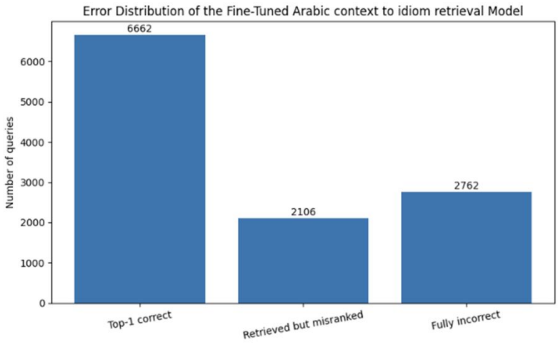

Figure 17: Error distribution across prediction outcomes.

A key observation is that a substantial portion of errors arises from ranking failures rather than retrieval failures. In many cases, the correct idiom is retrieved within the Top-5 candidates but is not ranked at the top position.

This indicates that the model successfully captures the overall semantic space but struggles with fine-grained discrimination between semantically similar idioms.

Fully incorrect predictions are typically associated with ambiguous Arabic inputs, weak contextual signals, or inherently difficult cross-lingual mappings.

Overall, the results suggest that improving ranking precision, rather than expanding retrieval coverage, is the primary direction for further performance gains.

8. *Similarity and Confidence Analysis*

We analyze the relationship between cosine similarity [5] scores and prediction correctness to assess model behavior and calibration.

Figure 18 shows that accuracy increases consistently with similarity score, indicating a strong positive correlation between similarity and prediction correctness. High-confidence predictions (similarity > 0.9) achieve very high accuracy, while low-similarity predictions are significantly less reliable.

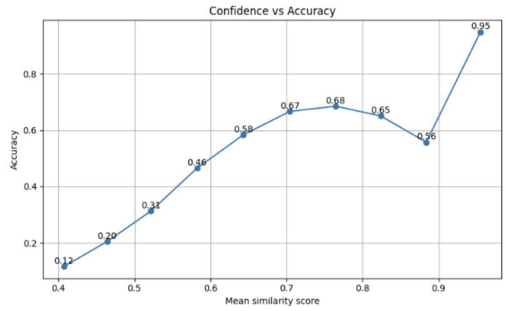

Figure 18: Accuracy as a function of similarity score.

Figure 19 further illustrates the distribution of similarity scores for correct and incorrect predictions. Correct predictions tend to have higher similarity values, while incorrect predictions are concentrated in lower score regions. However, there is a noticeable overlap between the two distributions.

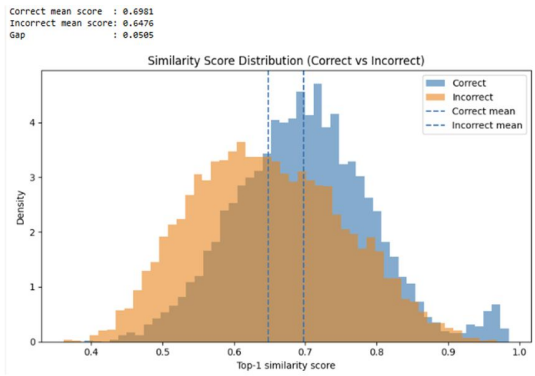

Figure 19: Similarity score distribution for correct and incorrect predictions.

This relatively small separation is also reflected in the mean similarity scores (0.698 for correct vs. 0.648 for incorrect), indicating that the embedding space is globally well-structured but locally dense.

The observed overlap suggests that many errors occur within semantically similar idiom clusters, rather than from complete semantic mismatches.

Overall, these findings show that cosine similarity provides a meaningful confidence signal but also highlight limitations in discriminative power when distinguishing between closely related idiomatic expressions.

9. *Threshold and Calibration Analysis*

We analyse model performance under varying similarity thresholds to evaluate its effectiveness as a confidence-based filtering mechanism.

Figure 20 shows that accuracy increases consistently as the similarity threshold becomes more restrictive. At lower thresholds, accuracy remains moderate due to the inclusion of ambiguous predictions, while higher thresholds (≥ 0.90) yield substantially higher accuracy, approaching near-perfect precision.

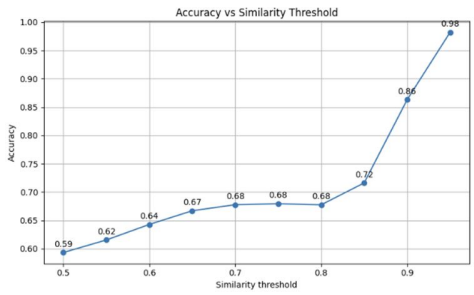

Figure 20: Accuracy as a function of similarity threshold.

This behavior reflects a clear trade-off between accuracy and coverage: stricter thresholds improve prediction reliability but reduce the number of accepted predictions.

These results indicate that similarity-based thresholding provides an effective calibration mechanism for practical deployment. By selecting an appropriate threshold, the system can balance precision and coverage, enable high-confidence predictions while deferring uncertain cases to fallback strategies.

10. *Recall@K Analysis*

We evaluate retrieval performance across different values of K to assess how often the correct idiom appears within the top-K candidates.

Figure 21 shows that recall increases rapidly for small values of K, with a large proportion of correct idioms retrieved within the Top-5 results. As K increases, the curve gradually saturates, indicating diminishing returns from retrieving additional candidates.

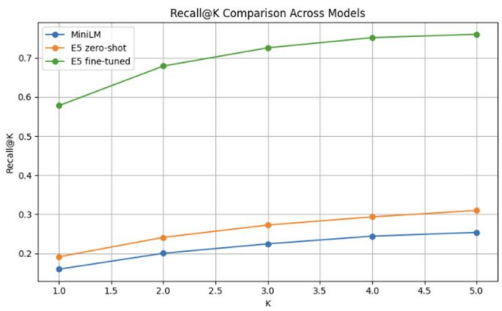

Figure 21: Recall@K performance across models.

This behavior suggests that the model is generally successful at identifying the correct semantic neighborhood, even when the correct idiom is not ranked at the top position.

In particular, the strong Recall@5 performance (0.760) indicates that the correct idiom is retrieved in most cases. However, the gap between Recall@1 and Recall@5 highlights persistent ranking limitations.

These findings reinforce earlier observations from the error analysis, confirming that the primary challenge lies not in retrieving relevant candidates, but in accurately ordering them within dense semantic clusters.

11. *Score Distribution Analysis*

Figure 22 compares the similarity score distributions across models. The fine-tuned E5 model produces a clear shift toward higher similarity values, indicating improved embedding alignment and stronger ranking confidence.

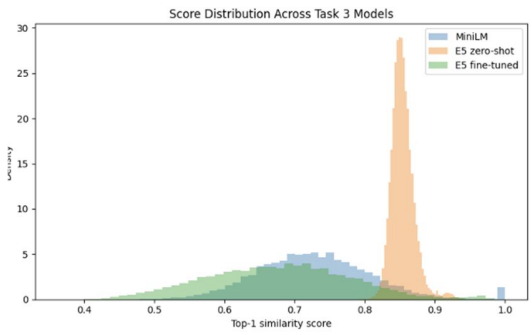

Figure 22: Similarity score distribution across models.

Compared to baseline models, the fine-tuned model exhibits better concentration of scores in higher similarity regions, reflecting improved semantic matching between Arabic contexts and idiomatic expressions.

This shift in distribution further confirms the effectiveness of task-specific fine-tuning in shaping a more discriminative embedding space.

12. *Qualitative Cross-Lingual Analysis*

To better understand model behavior, we present representative examples illustrating both successful predictions and typical failure cases.

The model performs well when the Arabic context provides clear semantic signals. For example:

- Arabic: "انتهى كل شيء في النهاية"
  Prediction: *come to an end*
- Arabic: "شارك همومك مع الآخرين لتخفيفها"
  Prediction: *a problem shared is a problem halved*
- Arabic: "لا توجد أي فرصة لحدوث ذلك"
  Prediction: *not a chance*

In these cases, the model successfully captures the underlying meaning and maps it to the appropriate idiomatic expression.

However, failure cases reveal consistent challenges:

- Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"
  Prediction: *please speak more slowly*
  Target: *slow down*

Here, the model retrieves a semantically related phrase rather than the canonical idiom, indicating sensitivity to surface-level similarity.

Across examples, successful predictions are typically associated with strong contextual signals and clear semantic intent, while errors often arise in ambiguous contexts or when multiple idioms share similar meanings.

Notably, incorrect predictions are often assigned high similarity scores, indicating overconfidence in dense semantic regions. This behavior aligns with earlier findings and highlights limitations in fine-grained ranking.

13. *Hard* Negative Analysis

Analysis of hard negative cases reveals that incorrect predictions are frequently semantically close alternatives to the correct idiom.

For example, idioms such as *in one fell swoop* vs. *at one stroke*, or *packed to the rafters* vs. *stuffed to the gills*, often appear within the same candidate set and are difficult to distinguish.

This indicates that the model struggles to differentiate between idioms within the same semantic cluster, where distinctions are subtle and context-dependent.

These findings suggest that incorporating hard negative mining during training could improve ranking precision by encouraging the model to better separate closely related idiomatic expressions.

14. Discussion

The results demonstrate that Arabic context-to-idiom retrieval is significantly more challenging than its monolingual counterpart due to cross-lingual semantic gaps and variation in idiomatic expression.

Unlike generation-based approaches, retrieval models rely on the structure of the embedding space to align meanings across languages. While multilingual models improve this alignment, they remain sensitive to subtle semantic differences and cultural variations in idiomatic usage.

The experimental results show that replacing the MiniLM [28] baseline with multilingual E5 leads to consistent improvements in retrieval performance. However, the gains from pretrained multilingual representations alone remain limited. The most substantial improvement is achieved through task-specific contrastive fine-tuning, which enables more precise alignment between Arabic contextual inputs and canonical English idioms.

Despite strong overall performance, the model continues to struggle with fine-grained semantic distinctions, particularly among idioms within the same semantic cluster. This highlights an inherent limitation of purely retrieval-based approaches for idiom understanding.

These findings suggest that future work may benefit from hybrid or generative approaches that can better capture subtle semantic nuances beyond nearest-neighbor retrieval.

15. Task Summary

In this task, we introduced a cross-lingual context-to-idiom retrieval benchmark, requiring the mapping of Arabic contextual inputs to canonical English idioms within a fixed candidate inventory.

The fine-tuned multilingual E5 model achieves the best performance, demonstrating strong capability in capturing cross-lingual semantic alignment. However, detailed analysis shows that most remaining errors arise from ranking ambiguity rather than retrieval failure.

This indicates that the primary challenge lies in distinguishing between semantically similar idioms within dense embedding regions.

Overall, the results highlight that effective cross-lingual idiom retrieval requires not only strong multilingual representations but also targeted training strategies that improve ranking precision.

These findings position cross-lingual idiom retrieval as a challenging and promising direction for multilingual semantic modeling, with clear opportunities for improvement through ranking optimization, hard negative mining, and enhanced contextual modeling.

These results highlight that cross-lingual idiom retrieval is not only a retrieval problem but also a semantic alignment challenge, where models must bridge figurative meaning across languages rather than rely on surface similarity.

VII. Discussion

The experimental results demonstrate that IdiomX supports a diverse range of NLP tasks related to idiom understanding, each requiring different modeling capabilities.

The findings highlight three key insights. First, idiom detection relies heavily on contextual modeling rather than surface-level lexical cues, confirming that idiomatic meaning cannot be inferred from isolated tokens. Second, context-to-idiom mapping is not purely a retrieval or generation problem, but requires precise semantic alignment between contextual meaning and canonical expressions. Third, hybrid approaches consistently outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

Across tasks, a consistent pattern emerges: while retrieval-based models are effective at identifying the correct semantic neighborhood, they often struggle with fine-grained ranking among semantically similar idioms. In contrast, generative models offer greater flexibility but may lack precision in producing canonical idiomatic forms. This suggests that idiom understanding benefits from a division of labor, where generative models provide contextual reasoning and retrieval mechanisms ensure discrete and accurate grounding.

In the cross-lingual setting (Task 3), these challenges become more pronounced due to semantic variation across languages. Although multilingual embeddings enable alignment between Arabic contexts and English idioms, performance remains limited by ambiguity within dense semantic clusters. This highlights the importance of improved ranking strategies and training methods that better capture subtle semantic distinctions.

Overall, the results position idiom understanding as a complex semantic task that requires integrating contextual reasoning, structured retrieval, and cross-lingual alignment within a unified framework.

VIII. Applications

The IdiomX dataset supports a wide range of natural language processing tasks related to figurative language understanding, including:

- idiom detection
- context-to-idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly well-suited for evaluating transformer-based models and large language models on non-compositional language, where meaning cannot be derived directly from surface forms.

IX. Dataset Availability

The IdiomX dataset is publicly available to support reproducibility and further research in idiomatic language understanding.

The dataset can be accessed through the following platforms:

Hugging Face:
https://huggingface.co/datasets/aymansharara/IdiomX

Kaggle:
https://www.kaggle.com/datasets/aymansharara/idiomx

The full data collection and enrichment pipeline is available at:

https://github.com/aymanshar/idiomx-dataset

Benchmark tasks, experiments, and evaluation scripts are provided at:

https://github.com/aymanshar/IdiomX

These resources include the dataset, preprocessing scripts, and task implementations, enabling straightforward integration and reproducibility.

X. Limitations

While IdiomX provides large-scale coverage and rich annotations, several limitations should be considered.

First, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Second, the Arabic data primarily reflects standard written Arabic and may not fully capture dialectal variation.

In addition, some generated examples may reflect model-preferred phrasing rather than naturally occurring usage frequencies. For the generation task, evaluation relies primarily on exact match metrics, which may not fully capture semantic equivalence between idiomatic expressions.

Future work includes incorporating human validation and extending the dataset to additional languages and dialects.

XI. Conclusion

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing broad coverage, rich semantic annotations, and aligned English–Arabic representations.

Through comprehensive benchmarking, we showed that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches for idiom detection, while hybrid generation–retrieval methods achieve the best performance for context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, retrieval-based models are effective at capturing general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

Overall, the results demonstrate that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

Future work includes extending IdiomX to additional languages using the same pipeline.

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